

Module 1: Introduction to Java Programming

Java Laboratory – Problem Statements

Module Overview

This module introduces students to the fundamentals of Java programming. Students will learn how to write, compile, and execute Java programs while understanding the concepts of variables, data types, operators, user input, type casting, and basic programming constructs.

Total Practice Problems: 15

Difficulty Distribution:

- Beginner: 6 Problems
 - Intermediate: 6 Problems
 - Advanced: 3 Problems
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Learning Objectives

After completing this module, students will be able to:

- Understand the structure of a Java program.
 - Write, compile, and execute Java applications.
 - Declare and initialize variables.
 - Use Java primitive data types.
 - Perform arithmetic and logical operations.
 - Accept user input using the Scanner class.
 - Apply type casting.
 - Solve simple computational problems using Java.
 - Display formatted output.
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Concepts Covered

- Java Program Structure
- main() Method
- Packages

- Variables
 - Constants
 - Primitive Data Types
 - Operators
 - Scanner Class
 - Type Casting
 - Escape Sequences
 - Comments
 - Formatted Output
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Laboratory Problem Statements

Problem 1: My First Java Program

Difficulty: Beginner

Problem Statement

Write a Java program to display the following information on the console:

- Your Name
- Roll Number
- Branch
- Semester
- College Name

Use separate lines for each item.

Concepts Covered

- Program Structure
 - `System.out.println()`
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Problem 2: Personal Information Card

Difficulty: Beginner

Problem Statement

Design a Java program that displays a formatted digital identity card containing:

- Name
- Father's Name
- Date of Birth
- Blood Group
- Mobile Number
- Email Address
- Address

Use escape sequences and proper formatting.

Problem 3: Exploring Java Data Types

Difficulty: Beginner

Problem Statement

Declare variables of the following data types:

- byte
- short
- int
- long
- float
- double
- char
- boolean

Initialize them with suitable values and display all values with their descriptions.

Problem 4: Simple Calculator

Difficulty: Beginner

Problem Statement

Accept two integers from the user and perform:

- Addition
- Subtraction
- Multiplication
- Division

- Modulus

Display the results neatly.

Problem 5: Area Calculator

Difficulty: Beginner

Problem Statement

Develop a Java program that accepts the radius of a circle and calculates:

- Diameter
- Circumference
- Area

Display the output up to two decimal places.

Problem 6: Temperature Converter

Difficulty: Beginner

Problem Statement

Accept temperature in Celsius and convert it into Fahrenheit.

Also convert Fahrenheit back into Celsius.

Intermediate Level Problems

Problem 7: Student Result Summary

Problem Statement

Accept the marks of five subjects.

Calculate:

- Total Marks
- Percentage
- Average Marks

Display all results in a formatted report.

Problem 8: Currency Breakdown

Problem Statement

Accept an amount from the user.

Display the minimum number of notes required using the following denominations:

- 500
- 200
- 100
- 50
- 20
- 10
- 5
- 2
- 1

(Assume all denominations are available.)

Problem 9: Swap Two Numbers

Problem Statement

Write a Java program to swap two numbers:

1. Using a third variable.
2. Without using a third variable.

Display the values before and after swapping.

Problem 10: ASCII Explorer

Problem Statement

Accept a character from the user and display:

- Character
- ASCII value

Also display the next five ASCII characters.

Problem 11: Type Casting Demonstration

Problem Statement

Write a Java program demonstrating:

- Implicit Type Casting
- Explicit Type Casting

Perform conversions among:

- int
- float
- double
- char

Display the original and converted values.

Problem 12: Employee Salary Summary

Problem Statement

Accept the following details:

- Employee Name
- Employee ID
- Basic Salary

Assume:

- HRA = 20% of Basic Salary
- DA = 10% of Basic Salary
- TA = 5% of Basic Salary

Calculate:

- Gross Salary

Display the salary slip in a professional format.

Advanced Level Problems

Problem 13: Electricity Bill Estimator

Problem Statement

Accept:

- Consumer Name
- Consumer Number
- Units Consumed

Assume a fixed rate per unit and a fixed service charge.

Calculate:

- Energy Charges
- Service Charges
- Total Bill

Display a formatted electricity bill.

Problem 14: Student Admission Form

Problem Statement

Design a Java application that accepts:

- Student Name
- Registration Number
- Course
- Branch
- Semester
- Mobile Number
- Email
- Percentage in Class XII

Display all information in a neatly formatted admission receipt.

Use different data types appropriately.

Problem 15: Smart Shopping Bill Generator

Problem Statement

Develop a Java application that accepts the following details for three products:

- Product Name
- Product Price
- Quantity Purchased

Calculate and display:

- Individual Item Total
- Subtotal
- GST (assume a fixed percentage)
- Grand Total

Display the result in the form of a shopping invoice.

Challenge Problems

1. Design a BMI (Body Mass Index) calculator.
 2. Create a loan EMI calculator using mathematical formulas.
 3. Develop a restaurant bill generator with GST and service charges.
 4. Create a travel expense calculator.
 5. Build a fuel consumption calculator for a vehicle.
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Real-World Applications

After completing this module, students should be able to develop simple applications such as:

- Student Information System
 - Salary Calculator
 - Shopping Bill Generator
 - Banking Transaction Summary
 - Electricity Bill Calculator
 - GST Invoice Generator
 - Temperature Converter
 - Unit Converter
 - Employee Information System
 - Personal Information Management
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Common Mistakes

Students should avoid the following common errors:

- Missing semicolons (;).
 - Incorrect class names or file names.
 - Declaring variables without initialization (when required).
 - Using the wrong data type.
 - Integer division when decimal results are expected.
 - Forgetting to import the `Scanner` class.
 - Incorrect use of escape sequences.
 - Using uninitialized variables.
 - Mixing `nextInt()` and `nextLine()` without handling the newline character.
 - Incorrect type casting leading to loss of precision.
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Viva Questions

1. What is the purpose of the JVM?
2. Differentiate between JDK, JRE, and JVM.
3. What is the significance of the `main()` method?
4. Explain the difference between `print()`, `println()`, and `printf()`.
5. What are Java primitive data types?
6. What is type casting? Differentiate between implicit and explicit type casting.
7. Why is the `Scanner` class used?

8. What are escape sequences? Give examples.
 9. What is the difference between variables and constants?
 10. Why is Java called a platform-independent language?
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Expected Learning Outcomes

Upon successful completion of Module 1, students will be able to:

- Develop and execute simple Java applications independently.
- Use variables, data types, and operators effectively.
- Accept user input and display formatted output.
- Apply type casting and basic arithmetic operations.
- Follow Java coding conventions and best practices.
- Build confidence for subsequent modules involving control structures, methods, arrays, and object-oriented programming.

Tips: Encourage students to compile and run each program from both the command line (`javac` and `java`) and an IDE such as Eclipse. This helps them understand the Java compilation process in addition to learning the language syntax.